1. The limit of a function

Exercise 1.5.

EXERCISE 1.5

2. Explain what it means to say that

$\lim_{x \to 1^-} f(x) = 3$ and $\lim_{x \to 1^+} f(x) = 7$.

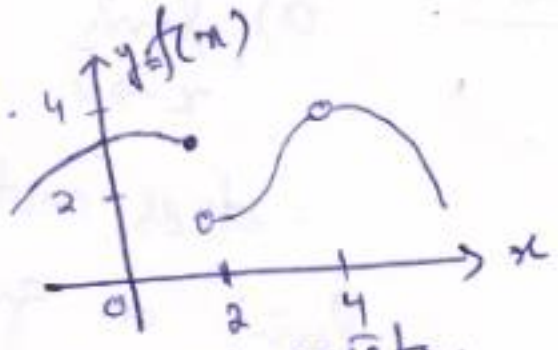
In this situation is it possible to say that $\lim_{x \to 1} f(x)$ exists? Explain.

Soln: $\lim_{x \to 1} f(x)$ does not exists b/c left- and right-hand limits are different.

4. Use the graph of "f" to state the value of each quantity. if it exists. If it does not exists, explain why.

a) $\lim_{x \to 2^-} f(x)$ b) $\lim_{x \to 2^+} f(x)$ c) $\lim_{x \to 2} f(x)$

d) $f(2)$, e) $\lim_{x \to 4} f(x)$ f) $f(4)$.



Soln: a) $\lim_{x \to 2^-} f(x) = 3$

b) $\lim_{x \to 2^+} f(x) = 1$

c) $\lim_{x \to 2} f(x)$ does not exists.

d) $f(2) = 3$, e) $\lim_{x \to 4} f(x) = 4$

f) $f(x)$ does not exists. ≡

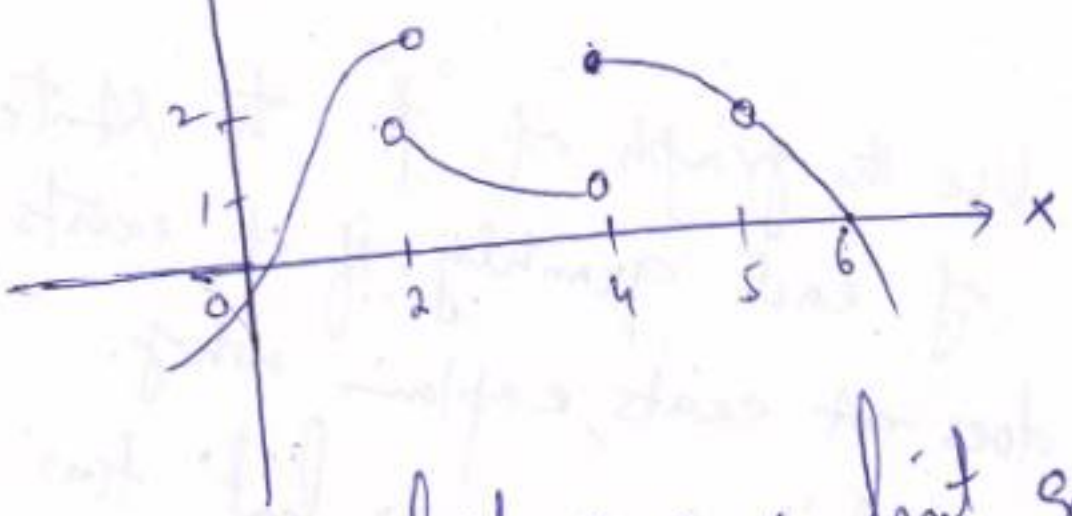
7. For the function g whose graph is shown, find a number "a" that satisfies the given description.

a) $\lim_{x \to a} g(x)$ does not exists but $g(a)$ is defined.

b) $\lim_{x \to a} g(x)$ exists but $g(a)$ is not defined.

c) $\lim_{x \to a^-} g(x)$ and $\lim_{x \to a^+} g(x)$ both exists but $\lim_{x \to a} g(x)$ does not exists.

d) $\lim_{x \to a^+} g(x) = g(a)$ but $\lim_{x \to a^-} g(x) \neq g(a)$.



Soln:

a) $\lim_{x \to 4^-} g(x) \neq \lim_{x \to 4^+} g(x)$, so $\lim_{x \to 4} g(x)$ does not exists. However, there is a point on graph representing $g(4)$. Thus $a = 4$ satisfies the given description.

b) $\lim_{x \to 5^-} g(x) = \lim_{x \to 5^+} g(x)$, so $\lim_{x \to 5} g(x)$ exists. But $g(5)$ is not defined. Thus,

$a = 5$ Satisfies the given descriptm.

c) At $a=4$, $\lim_{x\to 4^-} f(x)$ and $\lim_{x\to 4^+} f(x)$ exists but they are not equal, thus resulting that $\lim_{x\to 4} f(x)$ does not exists.

Also $\lim_{x\to 2^-} f(x)$, and $\lim_{x\to 2^+} f(x)$ exists, but $\lim_{x\to 2^-} f(x) \neq \lim_{x\to 2^+} f(x)$, and hence $\lim_{x\to 2} f(x)$ does not exists. So the point $a=2$ satisfies the given descriptm.

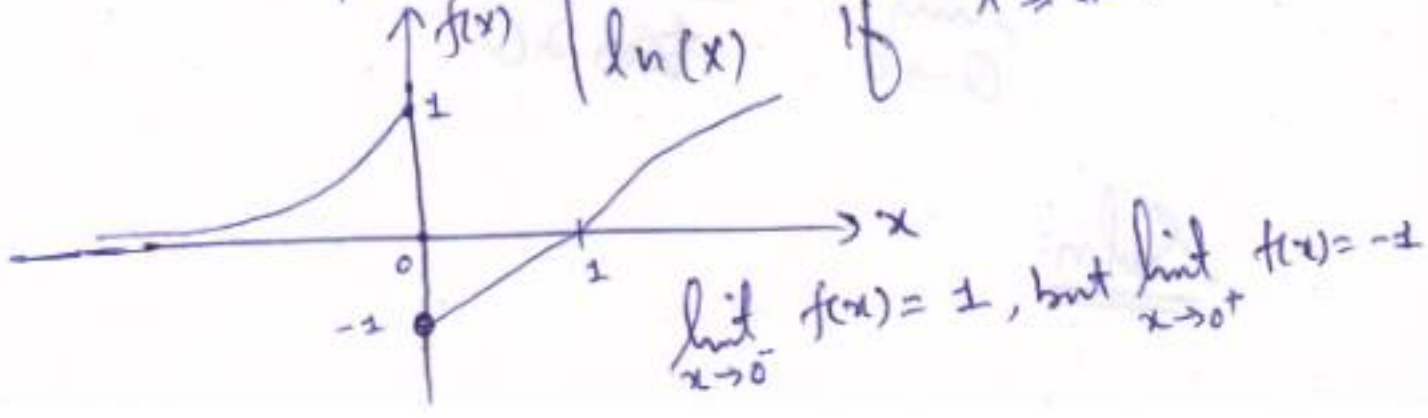
d) $\lim_{x\to 4^+} f(x) = f(4)$, but $\lim_{x\to 4^-} g(x) \neq g(4)$.

Thus $a=4$ satisfies the given descriptm.

II. Sketch graph of functin and use it to determine values of a for which $\lim_{x\to a} f(x)$ exists.

$$f(x) = \begin{cases} e^x & \text{if } x \leq 0 \\ x-1 & \text{if } 0 < x < 1 \\ \ln(x) & \text{if } x \geq 1. \end{cases}$$

Soln.



$\lim_{x\to 0^-} f(x) = 1$, but $\lim_{x\to 0^+} f(x) = -1$

So, $\lim_{x \to a} f(x)$ does not exist at $a = 0$.

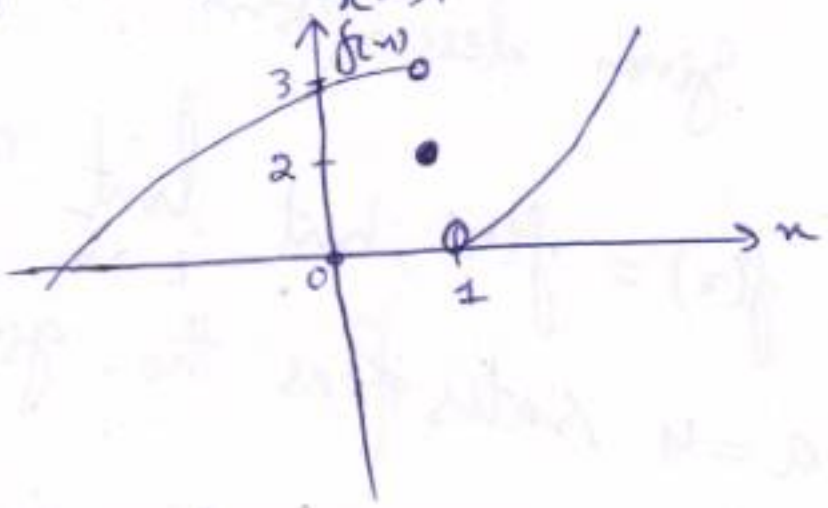
However, $\lim_{x \to a} f(x)$ exists for all other values of "a", thus, $\lim_{x \to a} f(x)$ exists for all "a" in

$(-\infty, 0) \cup (0, +\infty)$.

15: Sketch graph of an example of a function "f" that satisfies all given conditions:

$\lim_{x \to 1^-} f(x) = 3$, $\lim_{x \to 1^+} f(x) = 0$, $f(1) = 2$.

Soln.



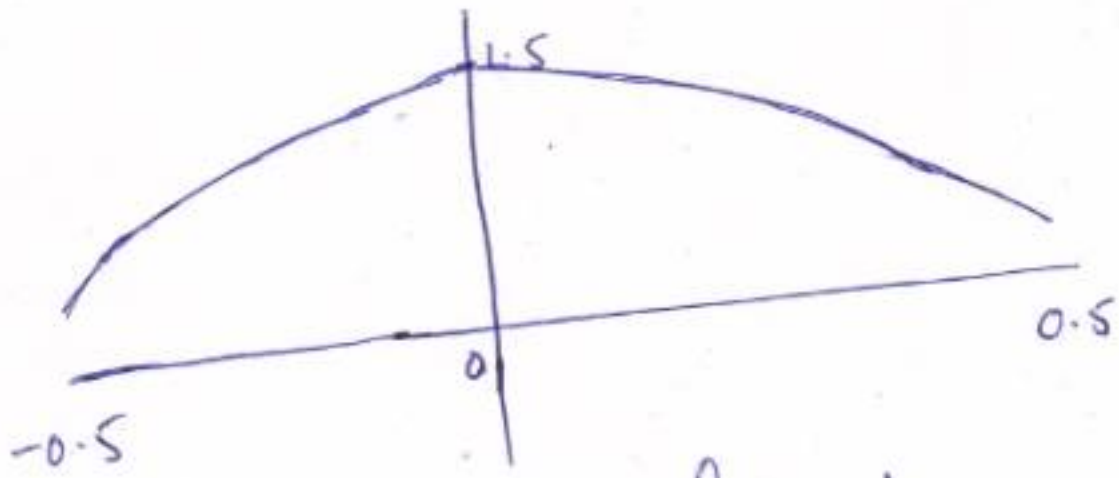
23: Use a table of values to estimate the value of limit. If possible use a graphing device to confirm your result graphically.

$$\lim_{\theta \to 0} \frac{\sin 3\theta}{\tan 2\theta}.$$

Soln:

θ	$f(\theta) = \frac{\sin 3\theta}{\tan 2\theta}$
± 0.1	1.457847
± 0.01	1.499575
± 0.001	1.499996
± 0.0001	1.500 000

It appears that $\lim_{\theta \to 0} \frac{\sin 3\theta}{\tan 2\theta} = 1.5$.



27. Determine the infinite limit.

$$\lim_{x \to 5^+} \frac{x+1}{x-5}.$$

Soln: $\lim_{x \to 5^+} \frac{x+1}{x-5} = +\infty$ b/c $x \to 5^+$.

28. Determine the infinite limit.

$$\lim_{x \to 5^-} \frac{x+1}{x-5}.$$

Soln: $\lim_{x \to 5^-} \frac{x+1}{x-5} = -\infty$ b/c $x \to 5^-$.

37. Find the vertical asymptote of the function

$$f(x) = \frac{x-1}{2x+4}.$$

Soln: The denominator of $f(x) = \frac{x-1}{2x+4}$ is equal to 0 when $x = -2$ (but not the numerator). So $x = -2$ is the vertical asymptote.

2. Calculating limit of a function by using laws of limit.

Exercise 1.6

Exercise 1.6:

1. Given that $\lim_{x\to 2} f(x) = 4$; $\lim_{x\to 2} g(x) = -2$, then find the limit that exists. If limit does not exists, explain why.

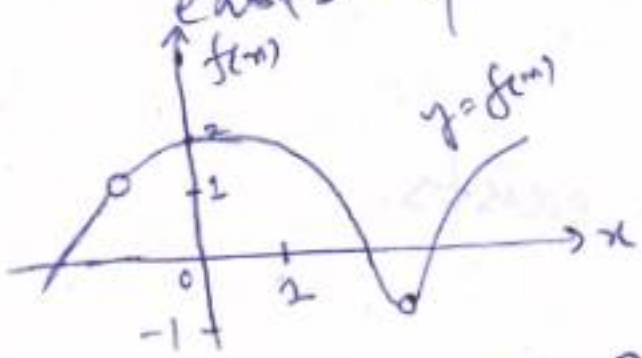
$$\lim_{x\to 2} [f(x) + 5g(x)].$$

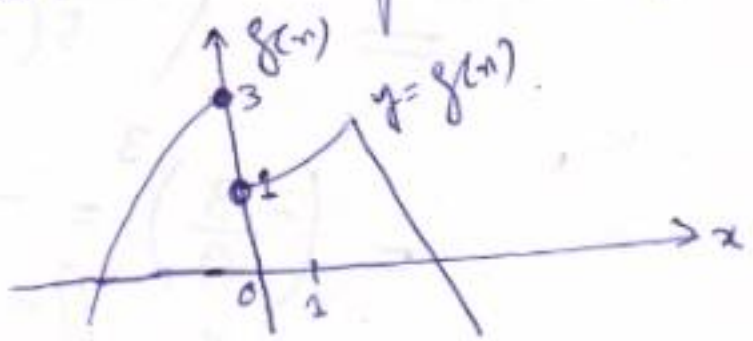
Soln: $\lim_{x\to 2} [f(x) + 5g(x)] \overset{\text{Law-I}}{=} \lim_{x\to 2} f(x) + \lim_{x\to 2} (5g(x))$

$$\overset{\text{law-III}}{=} \lim_{x\to 2} f(x) + 5\lim_{x\to 2} g(x)$$

$$= 4 + 5(-2) = 4 - 10 = -6.$$

2. The graph of f and g are given. Use them to evaluate each limit, if it exists. If limit does not exists, explain why.





$\lim_{x\to 2} [f(x) + g(x)]$.

Soln. $\lim_{x\to 2} [f(x) + g(x)] = \lim_{x\to 2} f(x) + \lim_{x\to 2} g(x) = -1 + 2 = 1.$

9. Evaluate the limit and justify each step by indicating appropriate limit law

$$\lim_{t\to -1}\left(\frac{2t^5-t^4}{5t^2+4}\right)^3.$$

Soln:
$$\lim_{t\to -1}\left(\frac{2t^5-t^4}{5t^2+4}\right)^3 \overset{\text{Law-6}}{=} \left(\lim_{t\to -1}\frac{2t^5-t^4}{5t^2+4}\right)^3$$

$$\overset{\text{law-5}}{=} \left(\frac{\lim_{t\to -1}(2t^5-t^4)}{\lim_{t\to -1}(5t^2+4)}\right)^3$$

$$\overset{\text{laws: 3,2, and 1}}{=} \left(\frac{2\lim_{t\to -1}t^5 - \lim_{t\to -1}t^4}{5\lim_{t\to -1}t^2 + \lim_{t\to -1}4}\right)^3$$

$$\overset{\text{laws: 10,8}}{=} \left(\frac{2(-1)^5-(-1)^4}{5(-1)^2+4}\right)^3$$

$$= \left(-\frac{3}{9}\right)^3 = -\frac{1}{27}.$$

11. Evaluate the limit if it exists.

$$\lim_{x\to -2}(3x-7).$$

Soln: $\lim_{x\to -2}(3x-7) = 3(-2)-7 = -6-7 = -13.$

23. Evaluate $\lim_{h\to 0} \dfrac{\sqrt{9+h}-3}{h}$.

Soln:

$$\lim_{h\to 0} \frac{\sqrt{9+h}-3}{h} \times \frac{\sqrt{9+h}+3}{\sqrt{9+h}+3}$$

$$= \lim_{h\to 0} \frac{(\sqrt{9+h})^2 - 3^2}{h(\sqrt{9+h}+3)}$$

$$= \lim_{h\to 0} \frac{9+h-9}{h(\sqrt{9+h}+3)} = \lim_{h\to 0} \frac{h}{h(\sqrt{9+h}+3)}$$

$$= \lim_{h\to 0} \frac{1}{\sqrt{9+h}+3} = \frac{1}{3+3} = \frac{1}{6}.$$

37. Use Squeeze theorem to show that

$$\lim_{x\to 0} x^2 \cos 20\pi x = 0.$$

Illustrate by graphing functions f, g and h where $f(x) = -x^2$, $g(x) = x^2 \cos 20\pi x$, and $h(x) = x^2$ on the same screen.

Soln: Let $f(x) = -x^2$, $g(x) = x^2 \cos 20\pi x$, $h(x) = x^2$.

Then, $-1 \le \cos 20\pi x \le 1 \Rightarrow -x^2 \le x^2 \cos 20\pi x \le x^2$

$\Rightarrow f(x) \le g(x) \le h(x)$.

Since, $\lim_{x\to 0} f(x) = \lim_{x\to 0} h(x)$, by Squeeze theorem, we have that $\lim_{x\to 0} g(x) = 0$



43: Find limit, if it exists. If the limit does not exists, explain why.

$$\lim_{x\to -4} \left(|x+4| - 2x\right).$$

Soln: $$|x+4| = \begin{cases} x+4 & \text{if } x+4 \geq 0 \\ -(x+4) & \text{if } x+4 < 0 \end{cases} = \begin{cases} x+4 & \text{if } x \geq -4 \\ -(x+4) & \text{if } x < -4 \end{cases}$$

Thus, $\lim_{x\to -4^+} \left(|x+4| - 2x\right) = \lim_{x\to -4^+} (x+4-2x) = \lim_{x\to -4^+} (-x+4)$

$= +4+4 = 8.$

$$\lim_{x\to -4^-} \left(|x+4| - 2x\right) = \lim_{x\to -4^-} \left(-(x+4) - 2x\right)$$

$$= \lim_{x\to -4^-} (-3x - 4) = 12 - 4 = 8.$$

Since, left and right limits are equal, so,

$$\lim_{x\to -4} \left(|x+4| - 2x\right) = 8.$$

3. The precise defination of limit.

Exercise 1.7.

Exercise 1.7:

Example-2: prove that $\lim_{x\to 3}(4x-5)=7$.

Soln: Let $\varepsilon>0$, be given. We aim to determine a number δ such that-

if $0<|x-3|<\delta$ then $|(4x-5)-7|<\varepsilon$.

But $|(4x-5)-7| = |4x-12| = |4(x-3)| = 4|x-3|$.

Therefore, we want a δ such that

if $0<|x-3|<\delta$ then $4|x-3|<\varepsilon$, that is,

if $0<|x-3|<\delta$ then $|x-3|<\frac{\varepsilon}{4}$. This suggests that $\delta=\frac{\varepsilon}{4}$.

To see if $\delta=\frac{\varepsilon}{4}$ works:

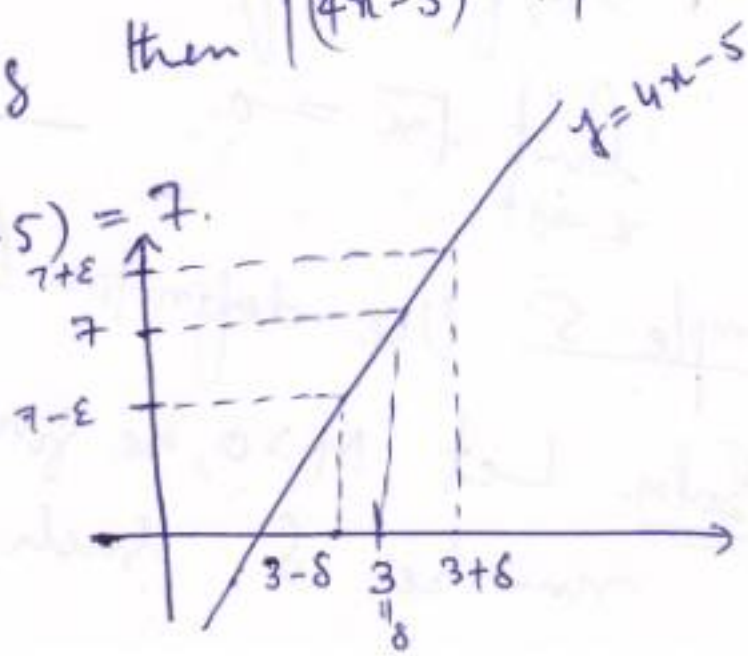
Given $\varepsilon>0$, choose $\delta=\frac{\varepsilon}{4}$.

If $0<|x-3|<\delta$, then

$$|(4x-5)-7| = |4x-12| = 4|x-3| = 4\left(\frac{\varepsilon}{4}\right)=\varepsilon.$$

Thus, if $0<|x-3|<\delta$ then $|(4x-5)-7|<\varepsilon$.

Therefore, $\lim_{x\to 3}(4x-5)=7$.



Example 4: Use definition 4 to prove that

$$\lim_{x \to 0^+} \sqrt{x} = 0.$$

Soln: Let $\varepsilon > 0$, be given. Here $a = 0$, $L = 0$. We aim to determine a number δ such that

if $0 < x < \delta$ then $|\sqrt{x} - 0| < \varepsilon$, that is,

if $0 < x < \delta$ then $\sqrt{x} < \varepsilon$, or

if $0 < x < \delta$ then $x < \varepsilon^2$. This suggests that $\delta = \varepsilon^2$.

To see if $\delta = \varepsilon^2$ works:

Given $\varepsilon > 0$, let $\delta = \varepsilon^2$. If $0 < x < \delta$, then

$$\sqrt{x} < \sqrt{\delta} = \sqrt{\varepsilon^2} = \varepsilon.$$

So, $|\sqrt{x} - 0| < \varepsilon$.

So, from definition 4, this shows that

$$\lim_{x \to 0^+} \sqrt{x} = 0. \quad \text{—}$$

Example 5: Use definition 6 to prove that $\lim_{x \to 0} \frac{1}{x^2} = \infty$.

Soln. Let $M > 0$, be given. We want to find a number δ such that

if $0<|x|<\delta$ then $\frac{1}{x^2} > M$.

But, $\frac{1}{x^2} > M \iff x^2 < \frac{1}{M} \iff \sqrt{x^2} < \sqrt{\frac{1}{M}} \iff$

$|x| < \frac{1}{\sqrt{M}}$.

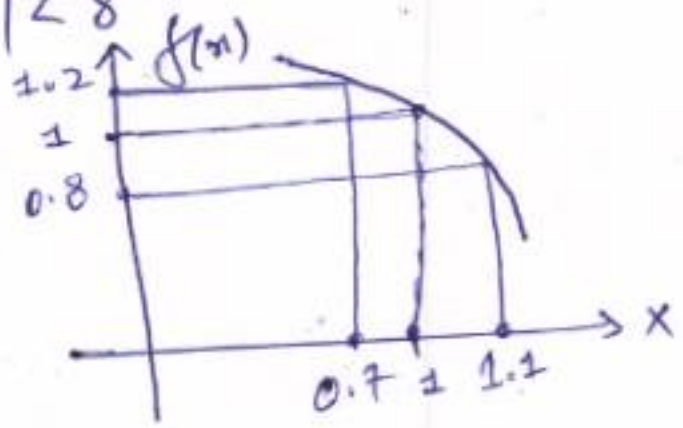
So, if we choose $\delta = \frac{1}{\sqrt{M}}$ and $0<|x|<\delta = \frac{1}{\sqrt{M}}$, then

$\frac{1}{x^2} > M$, this shows that $\frac{1}{x^2} \to \infty$ as $x \to 0$.

Exercise 1.7:

1: Use graph of f to find a number δ s.t

if $|x-1|<\delta$ then $|f(x)-1|<0.2$.



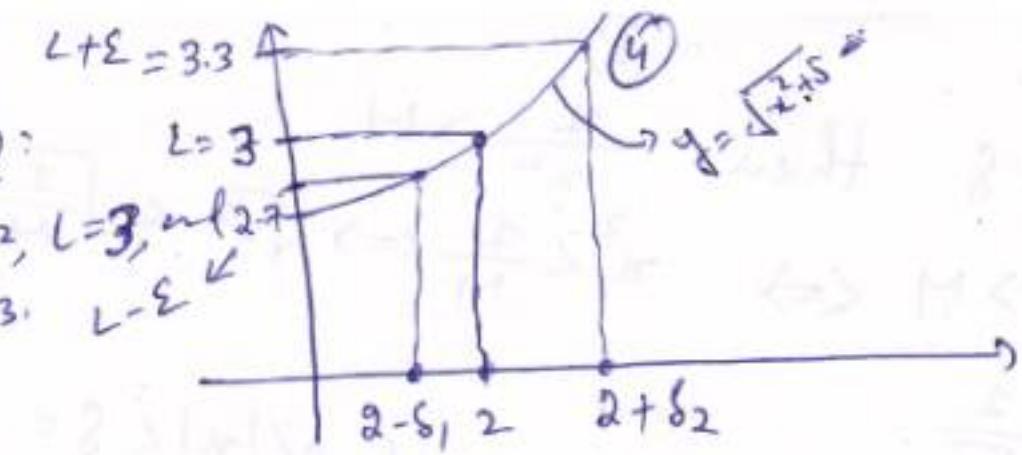
Soln:

If $|f(x)-1|<0.2$, then $-0.2 < f(x)-1 < 0.2 \Rightarrow$

$0.8 < f(x) < 1.2$. From the graph, we see that last inequality is true $0.7<x<1.1$, so we can choose $\delta = \min\{1-0.7, 1.1-1\} = \min\{0.3, 0.1\} = 0.1$.

5: Use a graph to find a number δ s.t

if $|x-2|<\delta$ then $|\sqrt{x^2+5} - 3| < 0.3$.



Soln:

Here, $a=2$, $L=3$, and $\varepsilon=0.3$.

Here, $\sqrt{x^2+5} = 2.7$

$\Rightarrow x^2+5 = 7.29$

$\Rightarrow x = \pm 1.513$

We take $\boxed{x = 1.513}$

So, $\delta_1 = 2 - 1.513$

$\boxed{\delta_1 = 0.487}$

$\sqrt{x^2+5} = 3.3$

$\Rightarrow x^2+5 = 10.89$

$\Rightarrow x = \pm 2.426$

We take $\boxed{x = 2.426}$

So, $\delta_2 = 2.426 - 2 = 0.426$

$\boxed{\delta_2 = 0.426}$

Since, $\delta = \min\{\delta_1, \delta_2\} = \{0.487, 0.426\} = 0.426$.

So, $\boxed{\delta = 0.426}$

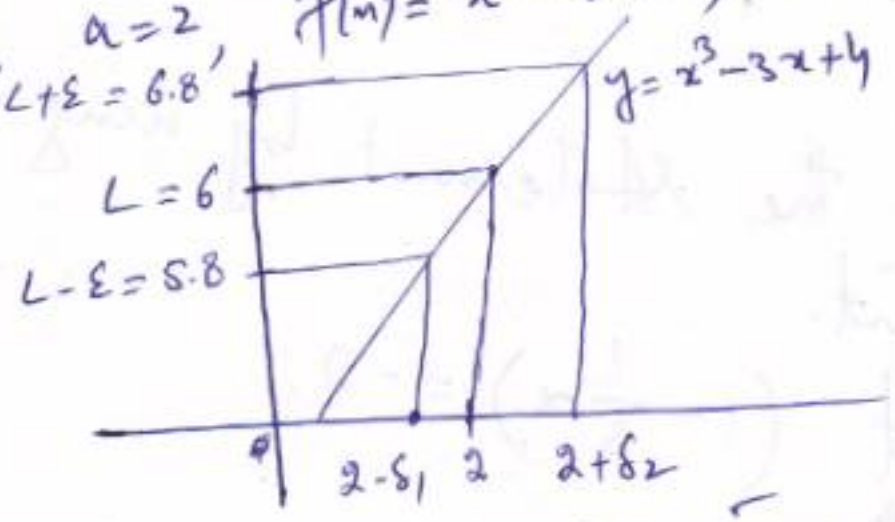
7. For the limit $\lim_{x \to 2} (x^3 - 3x + 4) = 6$ illustrate definition 2 by finding values of δ corresponding to $\varepsilon = 0.2$.

Solm:

If $|x-a|<\delta$ then $|f(x)-L|<\varepsilon$.

If $0<|x-2|<\delta$ then $|(x^3-3x+4)-6|<0.2$

Here, $a=2$, $f(x)=x^3-3x+4$, $L=6$, $\varepsilon=0.2$.



After simplification, we have that $x \approx 1.9774$, so that $\boxed{\delta_1 = 0.0226}$. And, when $x \approx 2.022$, we have that $\boxed{\delta_2 = 0.0219}$. Thus, $\boxed{\delta = 0.0219}$

15) Prove the statement using (ε, δ)-definition of limit and illustrate a diagram.

$$\lim_{x\to 4}\left(\frac{1}{2}x-1\right)=1.$$

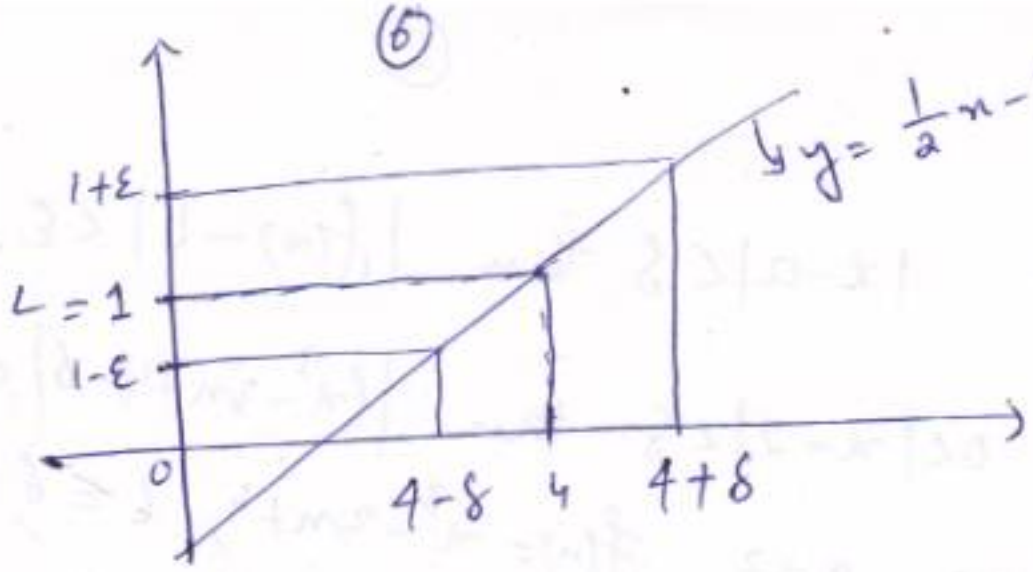
Solm: Given $\varepsilon>0$, we need $\delta>0$ such that

If $0<|x-4|<\delta$ then $\left|\left(\frac{1}{2}x-1\right)-1\right|<\varepsilon$.

But, $\left|\left(\frac{1}{2}x-1\right)-1\right|<\varepsilon \Leftrightarrow \left|\frac{1}{2}x-2\right|<\varepsilon \Leftrightarrow \frac{1}{2}|x-4|<\varepsilon \Leftrightarrow$

$|x-4|<2\varepsilon$, so we can choose $\boxed{\delta=2\varepsilon}$, then

$0<|x-4|<\delta \Rightarrow \left|\left(\frac{1}{2}x-1\right)-1\right|<\varepsilon$. Thus, $\lim_{x\to 4}\left(\frac{1}{2}x-1\right)=1$.



19: Prove the statement by using (δ,ε)-defn of limit.

$$\lim_{x \to 9} \left(1 - \frac{1}{3}x\right) = -2.$$

Soln: Given $\varepsilon > 0$, we need $\delta > 0$ such that

if $0 < |x-9| < \delta$ then $\left|\left(1 - \frac{1}{3}x\right) + 2\right| < \varepsilon$.

But, $\left|\left(1 - \frac{1}{3}x\right) + 2\right| < \varepsilon \iff \left|3 - \frac{1}{3}x\right| < \varepsilon \iff \left|-\frac{1}{3}\right||x-9| < \varepsilon$

$\iff |x-9| < 3\varepsilon$. So, if we choose $\delta = 3\varepsilon$, then

$0 < |x-9| < \delta \Rightarrow \left|\left(1 - \frac{1}{3}x\right) + 2\right| < \varepsilon$. Thus,

$\lim_{x \to 9} \left(1 - \frac{1}{3}x\right) = -2$, by definition of a limit.